

## Combined oxidative phosphorylation deficiency 54

[illegible]

HS FLA GAA AKE R SQMNS - - - - QTED - H - - ALAPVRNTIQLPTQPLNSE EWDK LKED LKENTGKTSF - - - ESWI ISQMAGCHSSID - - VAKSL  
 MM FAAGA AAKKR SHIGA - - - - NPQNQG - - HALPVRSSVQLPTKPLNSA EWDK LKEDFK - - - GKASF - - - EDFI ISQMARNCCSVD - - VAKSL  
 RN FAAGE AAKKR SHLSG - - - - NPQNQG - - HTLPVKSTVQLPTKPLNSE EWDK LKEDFK - - - GKASF - - - EDFV ISQMT RSCSSVD - - VAKSL  
 DR FSAGA AAKKR AEFLKHKSGLD TENDDT PVRVRDSPNMVLPDRPLTPSEW TNLKKDFS - - - RPELF - - - EKVM MENLVRANVDVN - - IAKSL  
 CE PTLNSPFP RPEI - - - - - - - - - - - - - - - - LEMSPKYQFL - - - - - - - - - - TTTTPQINGTTELQSM LSEVLR CNSSFFDSTV FNNV  
 DM P IAGH AHP RWLASQHKRRPQP GSLPADRLDQLRSDLFERRHE LSGA EWTEVRRSLVDGYKHINGHN - VDAV ILGVCSAPNQLS - - LAKSY

HS LAWVAAKNN - - - - - GIVSYDLL - - VKYLYLCVFH - - - - MQTSEVIDVFEIMKARYKTLEPRGYSLLIRGLIH - SDRWREALLLL  
MM LAWVAAKNN - - - - - GIVGYNLL - - VKYLYLCVFH - - - - KQTSEVIDVYEIMKAKYKSLESGGYTLLIRGLIH - SDRWRESLLLLL  
RN LAWVAAKNN - - - - - GIVGYNLL - - VKYLYLCVFH - - - - KQTSEVIDVYEIMKAKYKSLESGGYTLLIRGLIH - SDRWREALLLL  
DR LSYVVQ-DK - - - - - GTLSYEIL - - LRYLTICVRA - - - GHHSSEVFDAYDIMRSCFKSLDLGASALFIKGFSQ - TDRWKEAISML  
CE VKMVESQN-SFSRDFANLTAMRYIA - - - VVYHSINDEEARLTLKRLKDNIEKCKIHKMQ - - - - - EEHDKFRMSLD -  
DM VEFLRSRGSK - - - PNAATLGRLLRVYNAAHYHERPLSE - - - AEQSEILQICRSLQAHEHTLDASSCENVIHGLVATSGDDWQRGLPLL

190 200 210 220 230 240 250 260 270

|    |                                                                                                   |     |     |     |     |     |     |     |     |     |
|----|---------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| HS | EDIKKVITPSKKNYNDCIQGAL--LHQDVNTAWNLYQELLGHD--IVPMLETTLKAFFDFGKDIKDDNYSNKLLDILSYLRNNQLYPGESF       | 280 | 290 | 300 | 310 | 320 | 330 | 340 | 350 | 360 |
| MM | EDIKKVMVPSKKNYGDCIQGAL--LHQDVNTAWNLYQELIGHN--LIPPLETLKAFFDYGKDINDDHYSDKLLDILLYLRNNQLYPGESF        |     |     |     |     |     |     |     |     |     |
| RN | EDIKKVMVPSKKNYGDCIQGAL--LHQDVNVAWSLYQELVGHN--LIPLLETTLKAFFDHGKDMNDDQYSNQLLDILLYLRNNQLYPGESF       |     |     |     |     |     |     |     |     |     |
| DR | QDMKKALLPSPRNYGDVIAAGAV--LHGDSETSWKLYDELLELG--LVPQGDTWQCLFQSGI--TQRGHEDKLFDFVLSYMRDNQIYPEKPL      |     |     |     |     |     |     |     |     |     |
| CE | QDAAI PDFSKVMHKQDLVRCYF--RRELANKKWKT VKQLLSRYPKNIKL YEHI SAFL DAAKL-ENA-- --EPIDIMQVFMN YAQVDNPRF |     |     |     |     |     |     |     |     |     |
| DM | EMMKVTSTPSVAAYSALAEKAFSVATPEQELAWRLLEEMATAR--KPKCEVYLALLNRLAN-ETAQLPAQLSLLLQFLERHEILVSQRV         |     |     |     |     |     |     |     |     |     |

T>A (RCV003892100) 

HS AH--SIKTFWESVPGKQ---W---KGQFTTVRK--SGQCSGCGKTIESIQLSP EEEYEC LKGKIMRDVIDGGDQYRKTTTPQELK  
MM AH--SIKTFWESIPGRQ---W---KGQFTTIQK--SGQCSGCGRTIEPIHLSPEEYEF LKEKIMRDVIDGGDQYKKTTPQELK  
RN AH--SIKTFWESIPGRQ---W---KGQFTTIQK--SGQCSSCGRAIESIHLSP EEEYEF LKETIMRDVIDGGDQYKKTTPQELK  
DR ME--TIKAWFESLPDQK---W---RGKFSSVTP--SGDCRNCKALLEESIQLTEVEYEAQLKHNVMEKVIEGGDVFNKSNPEELK  
CE MEQTEFDEIFKQVAEACSAKVS VKSELDVKKITYISEKEGKILKEKCDKYVESMKEGFVKKEEYEEKIRQKVLRWME-----  
DM AI--RLQELSSQVP-QL-----L---HTRTT HLGGLGKCSQCQQLPVAISDEEFRQLSECF LERV LIRR DVFQRSTPEEVA

Sfs (RCV001824884) 

N>S (RCV001825014) 

● A > V (RCV002510555) 

A>D (RCV001825015) 

R>Q (RCV001824883) 

[illegible]

A>V (RCV001824882) 

|    |                               |       |     |     |     |        |       |        |       |        |       |        |       |       |       |      |      |      |     |      |     |   |     |   |     |      |     |      |   |     |   |   |     |   |
|----|-------------------------------|-------|-----|-----|-----|--------|-------|--------|-------|--------|-------|--------|-------|-------|-------|------|------|------|-----|------|-----|---|-----|---|-----|------|-----|------|---|-----|---|---|-----|---|
| HS | EDDPFLLYATLHSGNHCRFITRDLMRDHK | -     | -   | ACL | PD  | AKTQRL | FFKWQ | QGHQLA | IVN   | RF     | PGSKL | TFQR   | ILSYD | TVVQT | TGDS  | WHI  | PYDE | DL   |     |      |     |   |     |   |     |      |     |      |   |     |   |   |     |   |
| MM | EDDPFLLYATLNSGNHCKFITKDL      | LRD   | HK  | -   | -   | ACL    | PD    | ARTQRL | FFKWQ | QGHQLA | IMKG  | FQKSKL | TFQH  | ILSYD | TVVQR | TGDS | WHI  | PYDE | DL  |      |     |   |     |   |     |      |     |      |   |     |   |   |     |   |
| RN | EDDPFLLYATLNSGSHCKFITKDL      | LRD   | HK  | -   | -   | ACL    | PD    | ARAQRL | FFKWQ | QGHQLA | ITKG  | FLKSKL | TFQH  | ILSYD | TVVQT | TGDT | WHI  | PYDE | DL  |      |     |   |     |   |     |      |     |      |   |     |   |   |     |   |
| DR | EDDPYLLYAALNSGVHCKFLSR        | DLMRD | HK  | -   | -   | ACL    | PD    | SATRCL | FFKWQ | RGHQLV | ISHY  | TPGKR  | VRFER | ILSYD | TI    | IQT  | NGSS | WHI  | PYD | QKA  |     |   |     |   |     |      |     |      |   |     |   |   |     |   |
| CE | TDDLIIIRMALEFGPETSIVT         | NDKY  | TDH | RSA | I   | CNGD   | QKLE  | KIF    | DDFL  | TDASH  | RHFH  | -      | -     | -     | GRI   | ETR  | RNF  | NIR  | -   | IRNP | DNR | I | WIL | P | VMN | SE   |     |      |   |     |   |   |     |   |
| DM | HDDPFLLYATL                   | RS    | GQ  | ETD | F   | FSR    | DLMR  | SHA    | -     | -      | FHL   | G-     | TEL   | KPI   | FRR   | WQQE | HQ   | F    | SLV | TQT  | TGQ | I | IV  | K | EPV | RHRL | CAH | QVAD | T | WHV | P | Y | CAQ | Y |
|    |                               | 550   |     |     | 560 |        |       | 570    |       |        | 580   |        |       | 590   |       |      | 600  |      |     | 610  |     |   | 620 |   |     | 630  |     |      |   |     |   |   |     |   |

HS V - - E R - - - C S C E V P T K W L C L H Q K T - - - - -  
 MM V - - Q R - - - S S C E V P T K W L C L Q R K T P D P C - - - - -  
 RN V - - P R - - - S S C E V P T K W L C L Q R K T P A P C - - - - -  
 DR E - - D R - - - A S Y E V P Q K W L C L T R D - - - - -  
 CE G K L S R N I K R Y K V E V F R N S - - - - -  
 DM T P H P T - - - D S F E V P A N W L C I Q M K E Q T I P A T K T K T K T T R

640 650 660